



Headache Disorders

Learning objectives

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- Differentiate types of headache syndromes based on clinical features
- Recommend non-pharmacologic & pharmacologic treatment
- Construct individualized treatment regimens for acute and chronic management of headache syndromes
- Monitor headache treatment to ensure its safety and efficacy

Cont....

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- Headache is virtually **UNIVERSAL**
 - It is one of the top 10 presenting complaints in ambulatory medical care
- Headache can be symptom of a distinct pathologic process or can occur without an underlying cause
- **Primary** (tension-type, migraine, and cluster headache) and **Secondary** (symptom of organic disease)

Classification of Headaches

Primary

- No structural or metabolic abnormality
- No organic lesion in the background
- Treat the Headache!
- **Types**
 - Tension
 - Migraine
 - Cluster

Secondary/Symptomatic

- Structural or metabolic abnormality
- The headache is only a **symptom of another underlying disease**
- Treat the underlying disease!
- **Extracranial:** sinusitis, otitis media, glaucoma
- **Inracranial:** stroke(SAH), vasculitis, tumor, abscess, **meningitis**
- **Metabolic disorders:** CO poisoning

TABLE 14-1 Common Causes of Headache

Primary Headache		Secondary Headache	
Type	%	Type	%
Tension-type	69	Systemic infection	63
Migraine	16	Head injury	4
Idiopathic stabbing	2	Vascular disorders	1
Exertional	1	Subarachnoid hemorrhage	<1
Cluster	0.1	Brain tumor	0.1

PRIMARY HEADACHE

Type of Headache

Pain Location

Quality of Pain

Pain Severity

Tension type Headache



Sinus Headache



Migraine



Cluster Headache



MIGRAINE HEADACHE

Introduction

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“Migraine is a familial disorder characterized by **recurrent attacks of headache** widely variable in intensity, frequency and duration. Attacks are **commonly unilateral** and are usually associated with **anorexia, nausea and vomiting**”

-World Federation of Neurology

- Migraine is severe headache that interferes with **normal functioning** and if left untreated can last from 4 to 72 hrs

Facts about Migraine

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- One of the **most common complaint** encountered
 - Affects **10-20%** of the general population
- More than 2/3 of migraine sufferers either have never consulted a doctor or have stopped doing so
 - Migraine is underdiagnosed and undertreated
 - Greatly affect patients quality of life
- The WHO ranks migraine among world's most disabling medical illnesses

Epidemiology

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- Migraine is a ubiquitous neurologic disease that afflicts people of all ages
- It is highest between age 25-55 years
- Prevalence increases more rapidly in girls after puberty
 - Female: Male ratio is 2 : 1
 - Has been linked to menstruation
 - B/n 25-45 prevalence is highest in both gender

Migraine Triggers

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Food triggers

- Alcohol
- Caffeine/caffeine withdrawal
- Chocolate
- Fermented and pickled foods
- Monosodium glutamate (e.g., in Chinese food, seasoned salt, and instant foods)
- Nitrate-containing foods (e.g., processed meats)
- Saccharin/aspartame (e.g., diet foods or diet sodas)
- Tyramine-containing foods

Environmental triggers

- Glare or flickering lights
- High altitude
- Loud noises
- Strong smells and fumes
- Tobacco smoke
- Weather changes

Behavioral–physiologic triggers

- Excess or insufficient sleep
- Fatigue
- Menstruation, menopause
- Skipped meals
- Strenuous physical activity (e.g., prolonged overexertion)
- Stress or post-stress

Pathophysiology

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Vascular theory

- **Trigeminovascular system:** suggested that intracerebral vasoconstriction led to neural ischemia, which was followed by reflex extracranial vasodilation and pain due to release vasogenic peptides CGRP (calcitonin gene-related peptide.), neurokinin, sub P.

Unknown Trigger



Activation of trigeminal sensory nerves triggers

Release of vasogenic peptides



Vasodilation

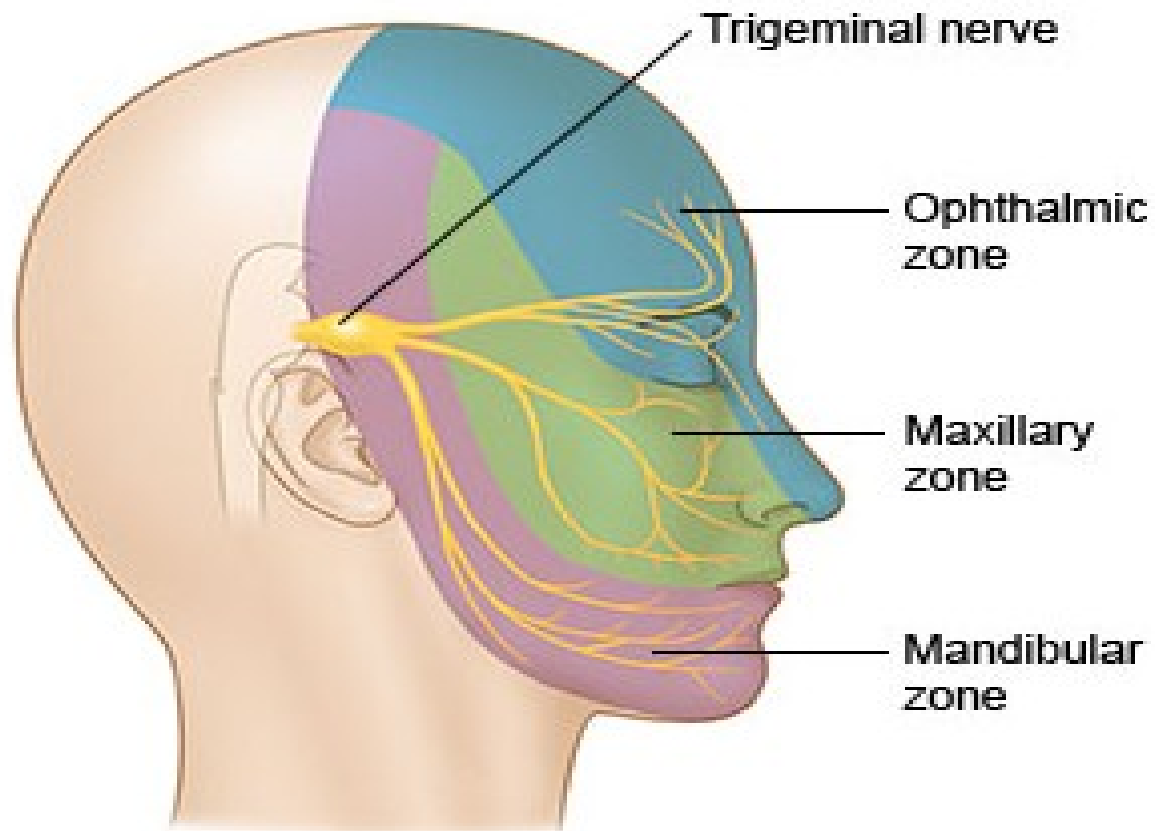


Promote neurogenic inflammation



Headache Pain

Trigeminal nerve



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Neuronal theory

- **Decreased serotonin levels** has been linked to migraine
 - Specific serotonin receptors are found in blood vessels of the brain (5HT-1B and 1D)

Neurovascular theory: New theory

- Defect or dysfunction in the activity of serotonin and excitatory neurotransmitter release in brainstem nuclei that modulate cerebral vascular tone and nociception

Classification of Migraine

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Migraine Without Aura	Migraine With Aura
No aura or Prodrome	Aura or Prodrome is present
Unilateral , throbbing headache that may be accompanied by nausea and vomiting	Unilateral throbbing headache that later becomes generalized
Moderate or severe intensity, aggravation by routine physical activity	Accounts 85% migraine
During headache, patient complains of phonophobia and photophobia	Patient complains of visual disturbances and may have mood variations

Clinical Presentation

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Migraine with aura has 4 phases

1. Prodrome

- Vague premonitory symptoms
 - begins 12-36 hours before the aura and headache
- Symptoms includes:
 - Yawning
 - Excitation
 - Depression
 - Lethargy
 - Craving or distaste for various foods

Duration: 15 to 20 min

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2. Aura

- Aura is a **warning or signal** before onset of headache
- Migraine aura symptoms include temporary visual or sensory disturbances that usually strike before other migraine symptoms — such as intense head pain, nausea, and sensitivity to light and sound

Symptoms

- **Flashing of lights**
- **Zig-zag lines**
- **Difficulty in focusing**

Duration : 15-30 min

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3. Active headache

- Headache is generally unilateral & throbbing and is associated with symptoms like:
 - Nausea
 - Vomiting
 - Photophobia
 - Phonophobia
 - Tinnitus

Duration is 4-72 hrs

4. Postdrome (Resolution phase)

Following headache, patient complains of

- Fatigue
- Depression
- Severe exhaustion
- Some patients feel unusually fresh

Duration: Few hours or up to 2 days

Diagnosis

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- Medical History
- Headache diary
- **Migraine triggers**
- Investigations (only to exclude secondary causes)
 - EEG
 - Brain CT or MRI
- **Common misdiagnoses for migraine:**
 - Sinus Headache (HA)
 - Stress HA

Treatment

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Goals of Migraine Treatment

- ▣ Reduce migraine frequency, severity, and disability
- ▣ Reduce headache-related distress and psychological symptoms
- ▣ Prevent headache and Improve quality of life

General Approach

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- Identification of triggers, meditation, relaxation training, psychotherapy
- **Abortive therapy:** to provide pain relief during an acute attack (**symptomatic relief**)
- **Preventive therapy:** to prevent or reduce the severity of recurrent headaches
 - ▣ Selection of initial therapy is based on headache severity

Acute Migraine Mgt (Non-Specific Agents)

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1. NSAIDs

- **Mild-moderate** migraine attacks or severe attacks that have been responsive in the past

Agents preferred: NSAIDS or acetaminophen plus caffeine

- **Acetaminophen alone is not proven effective**
- Pre-treatment with metoclopramide if N/V is severe

2. Opioids

- **Moderate-severe infrequent headaches** when:
 - When conventional agents are contraindicated or unresponsive to it
- Frequent use lead to **dependency and rebound headaches**

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3. Antiemetics

- Adjunct therapy for **N/V** associated with migraine & acute migraine medication such as ergotamine tartrate

Agents: metoclopramide administered 15 to 30 minutes before ingestion of oral abortive migraine medications

- Suppository preparation available if N/V is prominent

Specific agents (Ergot Alkaloids and Derivatives)

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- **Two agents:** Ergotamine tartrate and dihydroergotamine(DHE)
 - Considered for **moderate-severe attacks**
- **MOA:** Non-selective 5-HT₁ receptor agonists that **constrict intracranial blood vessels and inhibit inflammation**
- Both cause **venous & arterial constriction** (but ergotamine have more potent effect on arterioles)
- Ergotamine is available as oral, sublingual & rectal
 - Oral/rectal forms contain caffeine for increasing absorption & analgesia
 - **Ergotamine is most effective when administered early in migraine attack**
- DHE available as intranasal & parenteral(IM, IV, SC)

Side Effects

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- **Nausea and vomiting:** most common
 - Pretreatment with an antiemetic agent
- **Ergotism**
 - Severe peripheral ischemia manifested as cold, numbness, painful extremities, continuous paresthesias, diminished peripheral pulses
 - **Due to vasoconstrictive effect**(less with DHE)
 - In severe cases gangrene, hepatic necrosis, bowel & brain ischemia have been reported

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Note

- Avoid use of triptans within 24 of ergot use
- Avoid concurrent use with protease inhibitors(CYP450 inhibitor increase level of ergots)
 - Severe vasospasm can occur
- Contraindicated in renal, hepatic failure, coronary, cerebral or peripheral vascular disease, uncontrolled HTN, sepsis, pregnancy

Serotonin Receptor Agonists (Triptans)

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- **Agents:** **Sumatriptan**, zolmitriptan, naratriptan, rizatriptan, almotriptan, frovatriptan, and eletriptan
- **MOA:** selective agonists of $5HT_{1B}$ & $5HT_{1D}$

Place of therapy: 1st line for moderate-severe headache & used for rescue therapy when non-specific agents are ineffective

- Sumatriptan is the 1st agent approved with **fast onset**
 - SC (10min) vs. **oral (30min)** vs. intranasal (15min)
 - **Shorter half life:** headache recur in 30-40% pts within 24hr
 - Second dose must be given at the time of recurrence

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- 2nd generation agents provide better pharmacodynamics & pharmacokinetic profile than sumatriptan
 - Better oral absorption & longer half life
- If one triptan fail pts can be successfully switched to the other

Side Effects

- **Common side effects:** paresthesias, fatigue, dizziness, flushing, and somnolence
- **Local rxns:** Injection site rxn associated with SC
- **Contraindication:** IHD pts, uncontrolled hypertension, Cerebrovascular
 - postmenopausal women, men older than 40 years of age, and patients with multiple risk factors
- **Serotonin syndrome**
 - Concurrent use of MAOI, SSRI, SNRI

Dosing of Acute Migraine Therapies

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Drug	Dose	Usual Range/Comments
Analgesics		
Acetaminophen (Tylenol)	1,000 mg at onset; repeat every 4–6 hours as needed	Maximum daily dose is 4 g
Acetaminophen 250 mg/aspirin 250 mg/ caffeine 65 mg (Excedrin Migraine)	2 tablets at onset and every 6 hours	Available as nonprescription medication as Excedrin Migraine
Nonsteroidal Antiinflammatory Drugs		
Aspirin	500–1,000 mg every 4–6 hours	Maximum daily dose is 4 g
Ibuprofen (Motrin)	200–800 mg every 6 hours	Avoid doses >2.4 g/day
Naproxen sodium (Aleve, Anaprox)	550–825 mg at onset; can repeat 220 mg in 3–4 hours	Avoid doses >1.375 g/day
Diclofenac (Cataflam, Voltaren)	50–100 mg at onset; can repeat 50 mg in 8 hours	Avoid doses >150 mg/day
Ergotamine Tartrate		
Oral tablet (1 mg) with caffeine 100 mg (Cafergot)	2 mg at onset; then 1–2 mg every 30 minutes as needed	Maximum dose is 6 mg/day or 10 mg/wk; consider pretreatment with an antiemetic
Sublingual tablet (2 mg) (Ergomar)		
Rectal suppository (2 mg) with caffeine 100 mg (Cafergot, Migergot)	Insert 0.5 to 1 suppository at onset; repeat after 1 hour as needed	Maximum dose is 4 mg/day or 10 mg/wk; consider pretreatment with an antiemetic

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Dihydroergotamine

Injection 1 mg/mL (D.H.E. 45)	0.25–1 mg at onset IM, IV, or subcutaneous; repeat every hour as needed	Maximum dose is 3 mg/day or 6 mg/wk
Nasal spray 4 mg/mL (Migranal)	One spray (0.5 mg) in each nostril at onset; repeat sequence 15 minutes later (total dose is 2 mg or four sprays)	Maximum dose is 3 mg/day; prime sprayer four times before using; do not tilt head back or inhale through nose while spraying; discard open ampules after 8 hours

Serotonin Agonists (Triptans)

Sumatriptan (Imitrex)

Injection	6 mg subcutaneous at onset; can repeat after 1 hour if needed	Maximum daily dose is 12 mg
Oral tablets	25, 50, 85, or 100 mg at onset; can repeat after 2 hours if needed	Optimal dose is 50–100 mg; maximum daily dose is 200 mg; combination product with naproxen, 85/500 mg
Nasal spray	5, 10, or 20 mg at onset; can repeat after 2 hours if needed	Optimal dose is 20 mg; maximum daily dose is 40 mg; single-dose device delivering 5 or 20 mg; administer one spray in one nostril

Zolmitriptan (Zomig, Zomig-ZMT)

Oral tablets	2.5 or 5 mg at onset as regular or orally disintegrating tablet; can repeat after 2 hours if needed	Optimal dose is 2.5 mg; maximum dose is 10 mg/day
		Do not divide ODT dosage form
Nasal spray	5 mg (one spray) at onset; can repeat after 2 hours if needed	Maximum daily dose is 10 mg/day

Naratriptan (Amerge)

1 or 2.5 mg at onset; can repeat after 4 hours if needed	Optimal dose is 2.5 mg; maximum daily dose is 5 mg
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Rizatriptan (Maxalt, Maxalt-MLT)	5 or 10 mg at onset as regular or orally disintegrating tablet; can repeat after 2 hours if needed	Optimal dose is 10 mg; maximum daily dose is 30 mg; onset of effect is similar with standard and orally disintegrating tablets; use 5-mg dose (15 mg/day maximum) in patients receiving propranolol
Almotriptan (Axert)	6.25 or 12.5 mg at onset; can repeat after 2 hours if needed	Optimal dose is 12.5 mg; maximum daily dose is 25 mg
Frovatriptan (Frova)	2.5 or 5 mg at onset; can repeat in 2 hours if needed	Optimal dose 2.5–5 mg; maximum daily dose is 7.5 mg (three tablets)
Eletriptan (Relpax)	20 or 40 mg at onset; can repeat after 2 hours if needed	Maximum single dose is 40 mg; maximum daily dose is 80 mg
Miscellaneous		
Metoclopramide (Reglan)	10 mg IV at onset	Useful for acute relief in the office or emergency department setting
Prochlorperazine (Compazine)	10 mg IV or IM at onset	Useful for acute relief in the office or emergency department setting

Prophylactic Therapy

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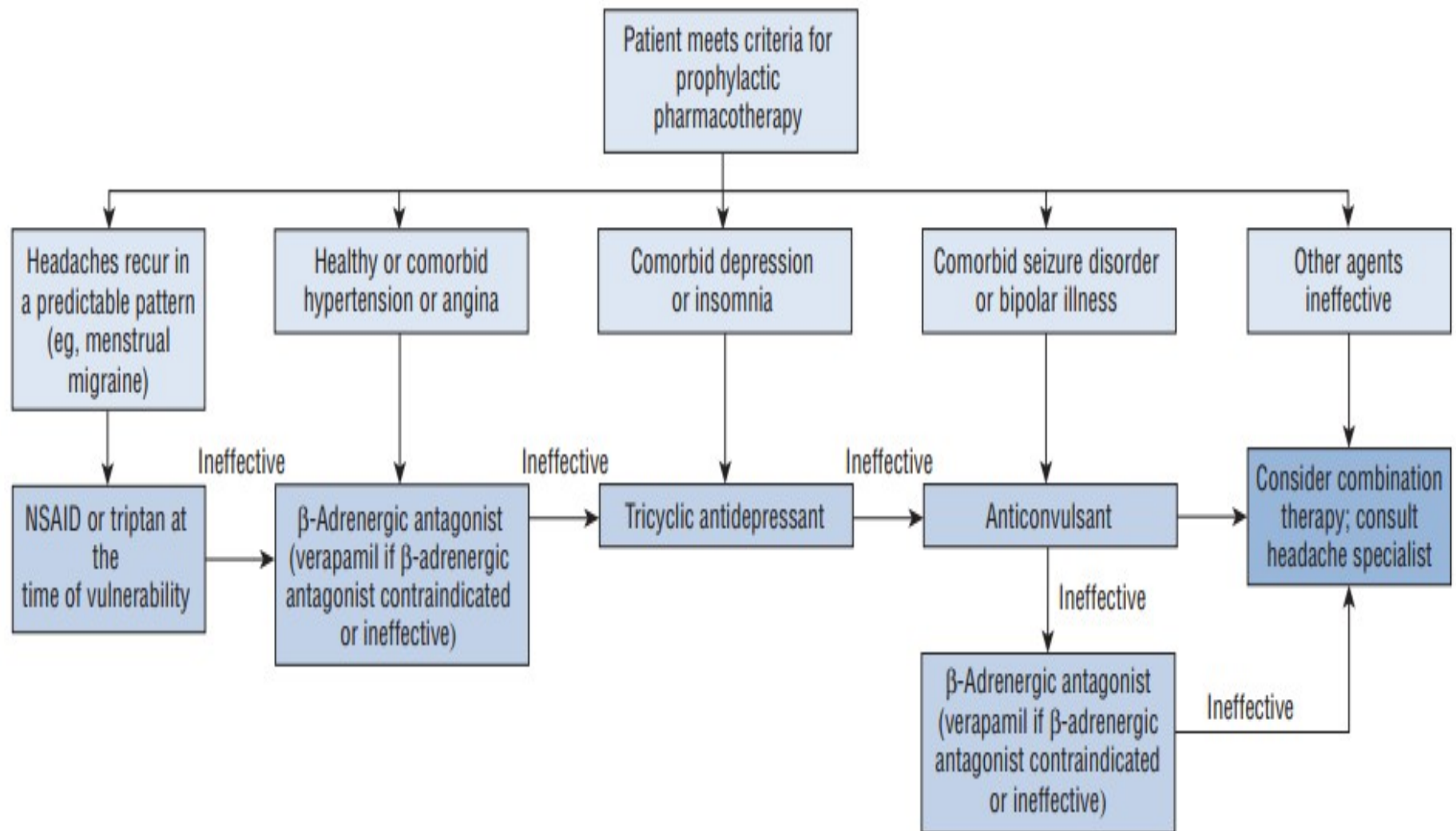
Preventive therapy should be considered when:

- Recurring migraines that produce significant disability despite acute therapy
- Frequent attacks requiring symptomatic medication ($> 2\times/\text{wk}$) with the risk of developing rebound headache
- Symptomatic therapies that are ineffective, contraindicated, or produce serious side effects
- Uncommon migraine type that cause profound disruption and/or risk of permanent neurologic injury
 - **hemiplegic migraine, basilar migraine,** and migraine with prolonged aura
- Patient preference to limit the number of attacks

Cont....

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- Preventive medication therapy reduces migraine frequency, severity, and headache-related distress
- **How do we choose prophylactic therapy?**
 - Efficacy
 - Concurrent disease states
 - Contraindications
 - Side effects
 - Compliance issues, and
 - Cost



Cont.....

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1. β -blockers

- The **most widely used drug for migraine prophylaxis**
- **Agents:** Propranolol (40 to 320 mg) , atenolol, and metoprolol
- **MOA** is unknown
 - Raise migraine threshold by modulating adrenergic or serotonergic neurotransmission in cortical or subcortical pathways
- Particularly useful in **patients with comorbid anxiety, hypertension, or angina**
- **Side effects:** bradycardia, and hypotension
- **Caution:** CHF, PVD, AV conduction disturbances, asthma, depression, and diabetes

2. Antidepressants: TCAs

- **Amitriptyline**: most widely studied and has demonstrated efficacy (10 to 300 mg)
- Other TCAs: doxepin, nortriptyline, and Imipramine also shown to be efficacious
- Anticholinergic & sedative side effects limit their use
- **Other**: orthostatic hypotension, cardiotoxicity, increase appetite, weight gain
- **SSRI**
 - Efficacy data are lacking only---used when co-morbid depression present

Cont....

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3. Anticonvulsants

- Effect may be associated with modulation of inhibitory, excitatory NT and ion channels
- **Divalproex sodium and Topiramate**, are particularly useful in migraineurs with comorbid seizures, anxiety, or bipolar

4. Calcium channel blockers

- 2nd or 3rd line for prophylaxis
- **Verapamil most widely used:**

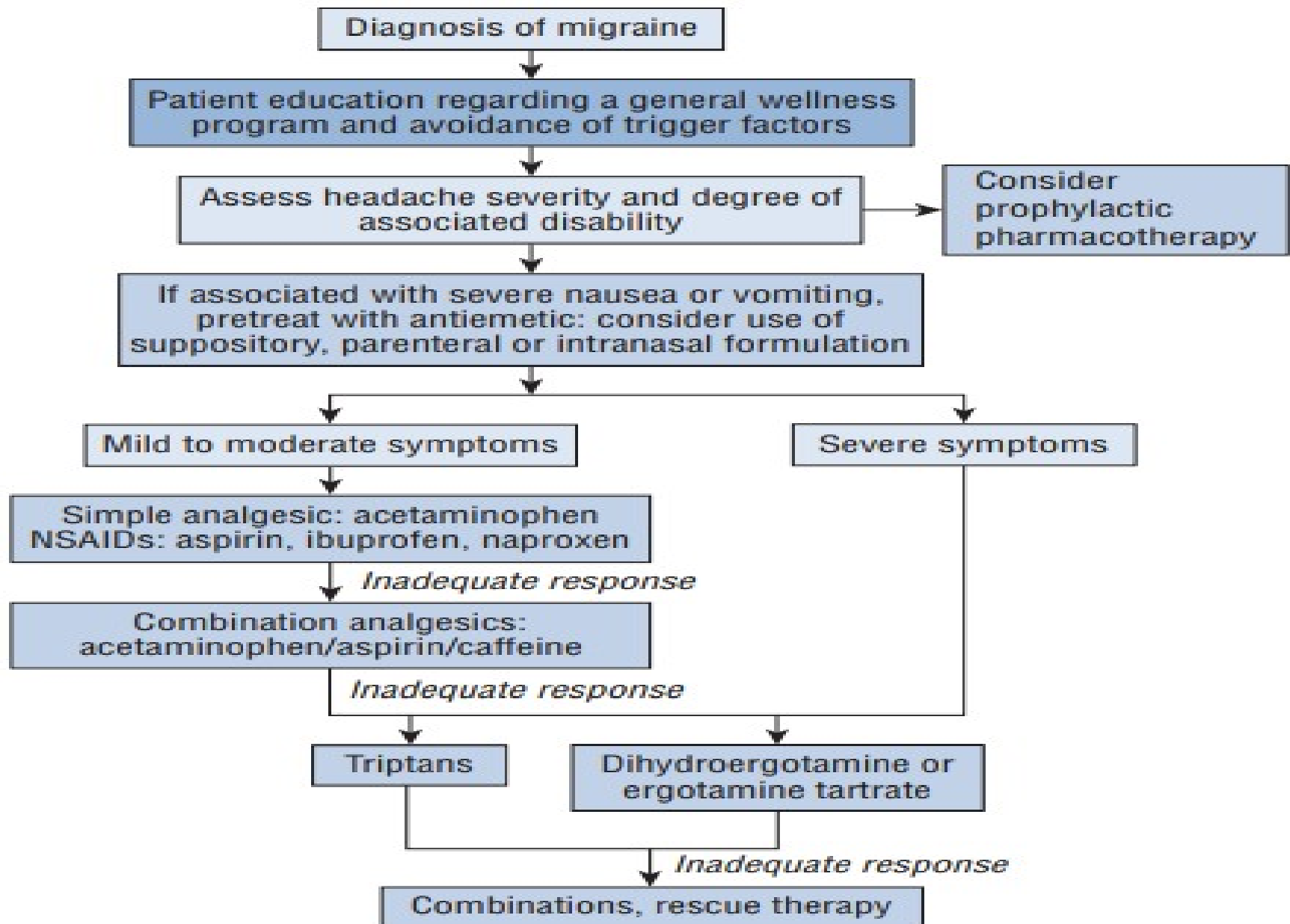
Side effects: Constipation, hypotension, bradycardia, AV block, and exacerbation of congestive heart failure

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NONSTEROIDAL ANTIINFLAMMATORY DRUGS

- are modestly effective for reducing the frequency, severity, and duration of migraine attacks, but**
- potential GI and renal toxicity limit daily or prolonged use.**
- may be used intermittently that recur in a predictable pattern (eg, menstrual migraine).**
- should be initiated 1 or 2 days before the time of headache and continued until vulnerability is passed.**
- The strongest is naproxen and weakest is aspirin.**



Case

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K.L., a 29-year-old woman, presents to the clinic with a 5-month history of left-sided pulsatile head pain recurring on a weekly basis. Her headaches are usually preceded by unformed flashes of light bilaterally and a sensation of light-headedness. The pain is always unilateral and is commonly associated with nausea, vomiting, and photophobia. The headache is not relieved by two tablets of either aspirin 325 mg or ibuprofen 200 mg and generally lasts all day unless she is able to lie in a dark room and sleep. The headaches usually interfere with her ability to continue work.

1. What should be the general approach to the treatment of K.L
2. What would be an appropriate drug of first choice for the treatment of K.L.'s acute headaches?
3. Is the patient fulfills the criteria for prophylaxis

TENSION-TYPE HEADACHE

Cont....

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Tension Type Headache

- Most common type (prevalence per year range 38% to 86%)
- More common in women than men
- Pain is usually mild to moderate, bilateral, non-pulsatile
- Premonitory symptoms and aura are absent
- Can be frequent episodic or chronic

Clinical Presentation

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- Mild to moderate in intensity
- Dull, non-pulsatile tightness or pressure.
- Bilateral pain is most common, location can vary
- The pain is classically described as having a “hatband” pattern.
- TTH is classified as either:
 - ▣ **Episodic:** (infrequent, fewer than one episode per month or frequent(episodes on 1-14 days/month) or
 - ▣ **Chronic:** frequent and duration (15 or more days/month)

Treatment

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- Majority of episodic TTH sufferers self medicate with OTC medications and do not consult a healthcare professional.
- Pharmacologic & non-pharmacologic treatments are available.
- **Analgesics and NSAIDs are the mainstay of therapy.**

Pharmacologic Therapy

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- **Simple analgesics** (alone or in combination with caffeine) and **NSAIDs** are effective for the acute treatment of **mild to moderate** TTH demonstrated efficacy
- The principles of **preventive treatment for TTH are similar to those for migraine headache**
- **TCAs** are prescribed most often for prophylaxis, but other drugs also can be selected after consideration of comorbid medical conditions and respective side effect profiles.
- venlafaxine, mirtazapine, gabapentin, and topiramate may also be effective

CLUSTER HEADACHE

Cont....

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- Severe form of the primary headache disorders. Also known as **'suicide headache'**
 - ❖ Characterized by attacks of excruciating ,deep, quick pain
 - ❖ Characterized by **severe strictly unilateral pain, lacrimation, nasal congestion, rhinorrhea, facial sweating or eyelid edema**
 - ❖ Occur periodically (weeks or months (ie, cluster periods), with active periods interrupted by spontaneous remissions(usually lasting months or years)

Epidemiology

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- The **most severe** of primary HA disorders
- Cluster HA can be **episodic** or **chronic**
- **Men are more likely than women**, onset generally occurs in those older than 20 years of age.

Pathophysiology

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- The etiologic & pathophysiologic mechanisms of cluster HA are not completely understood.
- The cyclic nature of attacks implicates a pathogenesis of **hypothalamic dysfunction** with resulting alterations in **circadian rhythms**
- Hypothalamus-regulated changes in **cortisol, prolactin, testosterone, GH, LH, endorphin, and melatonin** have been found during periods of cluster HA attack.

Clinical Presentation

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- Attacks occur in cluster periods lasting 2 wks to 3 mths, followed by long pain-free intervals
- Attacks occur suddenly, with pain peaking quickly after onset & generally lasting 15-180 min
- Auras are not present with cluster headaches
- The pain is excruciating, penetrating, and of a boring intensity in unilateral locations

Treatments

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- There's **no cure** for cluster headaches.
- Therapy for cluster headaches involves both **abortive** and **prophylactic** therapy
- The goal of treatment is to decrease the severity of pain, shorten the headache period and prevent the attacks

Abortive Therapy

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Oxygen

- The standard acute treatment of cluster headache is inhalation of 100% oxygen by facial mask at a rate of 7 to 10 L/min for 15 to 25 minutes
 - ▣ Causes a marked decrease in cerebral blood flow that is coincident with the reduced degree of pain in cluster headache
 - ▣ Repeated administration can be necessary because of recurrence,
- No side effects have been reported

Ergotamine Derivatives

- All forms of ergotamine have been used, but their role has been supplanted by the triptans
- Ergotamine tartrate also has provided effective relief of cluster HA attacks when administered sublingually or rectally
- Ergotamine can be an efficacious agent for **prophylactic** as well as **abortive therapy** of cluster headaches

Triptans

- The quick onset of subcutaneous and intranasal triptans make them safe and effective abortive agents for cluster HA
- **SC sumatriptan (6 mg) is the most effective agent.**
- Nasal sprays are less effective but may be better tolerated in some pts

- Adverse events reported in cluster HA pts are similar to those seen in migraine
- **Orally administered triptans** have limited use in cluster attacks because of **slow onset of action**;
- Oral zolmitriptan (10 mg), however, was beneficial in patients with episodic cluster headache with 60% experiencing mild or no pain at 30 minutes

Prophylactic Therapy

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Verapamil

- Verapamil, the preferred CCB for the prevention of cluster headaches, is effective in approximately 70% of patients.
- The beneficial effects of verapamil often appear after 1 week of therapy.
- A typical suggested dosage range is from 360 mg/day to 720 mg/day.

Lithium

- Lithium carbonate is effective for episodic and chronic cluster headache attacks and can be used in combination with verapamil
- The usual dose is 600 to 1,200 mg/day, with a suggested starting dose of 300 mg twice daily.

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Corticosteroids

- Corticosteroids are useful for inducing remission
- Therapy is initiated with 40 to 60 mg/day prednisone and tapered over $\approx$ 3 wks.
- Relief appears within 1 to 2 days of initiating therapy.
- To avoid steroid-induced complications, long-term use is not recommended.
- HA can recur when therapy is tapered or discontinued

Cases

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1. M.R., a 29-year-old woman, has throbbing right-sided headaches. She experiences nausea, phonophobia, and photophobia with these headaches but no aura. She usually has headaches twice a month. She has hypertension and morbid obesity. She takes hydrochlorothiazide 25 mg/day orally for hypertension, which is the only medication she is taking. Which medication is best for prophylaxis of her headaches?

A. Propranolol. B. Indomethacin. C. Topiramate. D. Sumatriptan.

THE END